Homework

HW11 — A face and its height — 50 min.

In an orthonormal frame with origin O, consider the points $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Prove that the triangle ABC is equilateral and calculate its area.
2. Show that the vector $\vec{n} = (1;1;1)$ is normal to the plane (ABC).
3. Let H be the orthogonal projection of O onto (ABC). Justify that its coordinates are of the form $(t;t;t)$. Find t using $\overrightarrow{AH} \cdot \vec{n} = 0$.
4. Calculate OH, then the volume of the tetrahedron $OABC$ taking ABC as the base. Check this volume by taking OAB as the base.